Bahria University Lahore Campus

Department of Computer Science

Course Code/number	**CSC – 210**
Course Title/Name	**Object Oriented Programming**
Credit Hours/Contact hours	3
Lab Credit Hours/Contact hours	1
Degree Program	Bachelor of Science in Computer Science (BS – CS)
Prerequisites or Corequisites	CSC-113 - Computer Programming
Assessment Methods and Weightage	Quizzes/Viva/Participation 10 Assignments/Projects 20 Mid-Term Examination 20 Final Examination 50 **Total** **100**
Textbook (or **Laboratory Manual** for Laboratory Courses)	[A] H. M. Deitel "C++ How to Program", 5th Ed., Prentice Hall, 2005. ISBN 0-13-185757-6. [B] R. Lafore, Object-Oriented Programming in C++, 4th Edition, Sams publishing, 2002.
Reference Material (With Edition)	R1. Tony Gaddis, "Starting out with C++", 6th Edition, Pearson, 2011. R2. Handouts. R3. Herb Schildt, "C++ The Complete Reference", 4th Ed, McGrawHill, 2002. R4. R. Lafore, Object-Oriented Programming in C++, 4th Edition, Sams publishing, 2002.
Web Resources/ URL (if any)	-

<table>
<tr><td>Instructor Name/Cluster Head/Subject Expert</td><td>Instructor Name: Ms. Fatima Zulfiqar Designation: Lecturer Status ☑ Regular □ Visiting Cluster Head Name: Dr. Iram Noreen</td></tr>
<tr><td>Course Aims and Objectives</td><td>• This is an introductory course on object-oriented programming, which is designed to develop understanding of fundamental concepts of object-oriented programming.
• The course covers several basic and advanced object-oriented concepts including classes, objects, inheritance, polymorphic, composition, encapsulation, templates etc.
• The course aims to illustrate the object-oriented concepts and develop solutions using C++ and a little bit of JAVA language, their design principles, and tools.</td></tr>
<tr><td>Learning Outcomes:</td><td>After the successful completion of course, the students will be able to:
<table>
<tr><td>CLO1</td><td>Develop an understanding of the underlying concepts of object-oriented paradigm</td><td>PLO1: Academic Education</td><td>Cognitive</td><td>2</td></tr>
<tr><td>CLO2</td><td>Apply object-oriented concepts in solving computing problems</td><td>PLO2: Knowledge for Solving Computing Problems</td><td>Cognitive</td><td>3</td></tr>
<tr><td>CLO3</td><td>Design and implement object-oriented solutions for small real world problems including single/multiple objects</td><td>PLO4: Design and Development of Solutions</td><td>Cognitive</td><td>4</td></tr>
<tr><td>CLO4</td><td>Evaluate the workflow of an object-oriented solution including error handling</td><td>PLO4: Design and Development of Solutions</td><td>Cognitive</td><td>5</td></tr>
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<tr><td>Course Outcomes</td><td>Along with many others following are fundamental learning outcomes for students that are expected through this course
• Practical understanding of Abstract Data Types (classes) to build small units to pack the data inside the code
• How to create classes and objects in C++; and various related constructs: access specifiers, constructor, destructor, const, static, etc.
• Difference between static and dynamic polymorphism and C++ syntax to achieve them. Several types of inheritance: linear, multiple, virtual (to resolve diamond shaped hierarchy)
• Creating generic (type independent) functions and classes
• Other C++ specific stuff e.g. I/O streams, exceptions handling</td></tr>
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Course Description/Catalogue	This course starts with the basic concepts (e.g., data abstraction and encapsulation) and definitions in Object Oriented paradigm and continues to teach coding these concepts in C++. It covers writing code for simple class aka ADT (Abstract Data Type) aka UDT (User-defined Data Type), adding constructor, destructor, and other basic methods (setter, getter, etc.). It also introduces type modifiers (const and static) available in C++. It also shows how to create Array and Pointer of instances of ADT or class, and other relations (composition, friend, inheritance). It also teaches C++ facilities of operator overloading, class templates, overriding, stream I/O and exception handling.			
Lecture Plan (16 Weeks)	**Week # and Date**	**Lecture / Hour #**	**Topic to be covered**	**Reference Text**
	1	1	**Course Introduction** Overview of Structured Programming: Focus (Pointer, Arrays, Dynamic Memory Allocation, Structures)	R1: Reading Material
		2	Overview of Structured Programming Continued	R1- (Reading Material)
	2	3	Intro to Object Oriented Concepts and Terminology: Real world examples	R2 R2
		4	**Define a Class** Access modifiers: private and public only. First Member function. Data Abstraction through Encapsulation. Setter/Mutator and Getter/Accessor methods. How the function knows which object invoked him? Answer will be given in next week.	A-(Ch-3 (3.1~3.6))
	3	5	**Revision of Function Overloading & Default Arguments.** Two Special Method: Constructor & Destructor. Constructor with No argument (Default Constructor). Constructor with Default arguments Calling sequence of Constructor & Destructor for multiple objects. Pointer as data member Importance of destructors Pointer /Reference to objects, Passing objects to functions by reference Pointer this	A-(Ch-3 (3.7~3.11)) A-(Chapter-9) A-(Chapter-10.5, 10.6)

		6	Passing objects to functions by value, Default Member-wise copy (Assignment & Initialization), Problem of Member-wise copy & its solution: Copy Constructor, Calling sequence of Constructor & Destructor: when objects are passed by value/reference	R2
	4	7	Preventing changes in data members from a method: const method. Constant data members Static functions Static data members const and static objects Calling sequence of Constructor & Destructor for const and static objects	A-(Chapter 10.2, 10.7)
		8	Object as data member (Composition), Aggregation. Cascading calls with and without this pointer. Nameless objects Array of objects;	A-(Chapter 10.3) R2
	5	9	**Operator Overloading.** Binary Operator receiving Instance of class as 1st operand	A-(Chapter 11.7)
		10	Overloading Unary Operator: as member, as non-member Unusual Operators: ++, --,	A-(Chapter 11.11)
	6	11	[] (set & get, both versions), type-cast, Parentheses	A-(Chapter 11.8, 11.9)
		12	**Friend functions** (efficient but shake the concept of encapsulation). Declaring a global function as friend of a class. Declaring member function of a class as friend of another class Friend class (an easy but more un-secure way);	A-(Chapter 10.4)
	7	13	Operator NOT receiving Instance of class as 1st operand Overloading Binary Operators (Stream Insertion/Extraction) for I/O stream	A-(Chapter 11.5)
		14	Other unusual operators: comma, arrow, new, delete, delete [] etc.	R2
	8	15	Review of the course for Mid Term Exam	
		16	Review of the course for Mid Term Exam	
	Mid Term Exam			

	9	17	**Inheritance** In theory, what is inheritance? is-A Relationship: public Inheritance protected data member protected and private inheritance	A-(Chapter 12.1~12.4) R2
		18	**Multilevel Inheritance** Direct and Indirect Base Class Calling of Constructor and Destructor for Derived Class Objects. Explicit call to the constructor of Base class from Derived class;	A-(Chapter 12.5, 12.6) Case Study-A
	10	19	Review of Simple Inheritance, Multi-Level Inheritance, Multiple Inheritance A review of OO relationships terminologies: Aggregation, composition, generalization (is-A), knows-a.	A-(Chapter 12) Case Study-B
		20	**Polymorphism** Overriding base-class members in derived class; Virtual functions and Dynamic binding. Concept of v-Pointer and v-Table	A-(Chapter 13) R1-Reading Material Case Study-B
	11	21	Pure virtual functions and abstract class. Defining a pure virtual destructor	R1-Reading Material
		22	Detail discussion on v-table structure, virtual constructor, object cloning	R1-Reading Material
	12	23	Some tricky things in pointer type casting and applying them on different topics studied so far.	R1-Reading Material
		24	Diamond inheritance Virtual inheritance	R1-Reading Material
	13	25	**File Handling** C++ Streams, Members and Manipulators of Streams. File Handling using Streams	R1-Reading Material A-(Chapter Chapter15)
		26	Access Techniques: Sequential, Direct, and Random-Access Files. Input/Output of Object from/to File (binary/ text mode);	R1-Reading Material A-(Chapter Chapter17)
	14	27	**Templates** <u>Function Template.</u> Overloading of Function Template. Specialized of Template Function	A-(Chapter 14.2,14.3)

		28	Class Template. Specialized method of Template Class. Complete Specialized Template Class	A-(Chapter 14.4, 14.5)
	15	29	Friendship and Inheritance with Templates	A-(Chapter 14.6, 14.7, 14.8)
		30	**Exception Handling** What is Exception? Error vs. Exception. Evolution of Exception Handling: exit, abort, assert, new keywords; try, catch, throw Unhandled Exception. Propagation of Exception and its advantage	A-(Chapter 16)
	16	31	Overall Review of Course for Final Exam	
		32	Overall Review of Course for Final Exam	
	Final Term Exam			

Semester Calendar for Assignments/Quizzes/Project

Assignments/ Projects and quizzes Plan	Week #	Assignment No.	Quiz No.	Project Title	Assignment/Project Due Date	Result Date of Assignment/Project/Quiz
	1					
	2					
	3		Quiz-1			
	4	Assignment-1				Quiz-1
	5		Quiz-2		Assignment-1	Assignment-1
	6	Assignment-2				Quiz-2
	7		Quiz-3		Assignment-2	Assignment-2
	8					Quiz-3
	10	Assignment-3				
	11		Quiz-4	Proposal Announced	Assignment-3	
	12			Design Announce	Project Proposal	Quiz-4, Assignment-3
	13	Assignment-4		Development	Project Design	
	14		Quiz-5		Assignment-4	Quiz-5
	15				Project Development	Assignment-4

Instructor Name: **Fatima Zulfiqar**

Instructor Signature: _____

Date: **20th February 2023**